

Sri Lanka Institute of Information Technology

IE3151 – Information Systems Project Management

Final Project Report

Final Report Submission: Project Execution, Monitoring and Control, Closure, Performance Domains, Project Management Principles and Lessons Learned

Client organisation: **Saara Ketha Harvest Hub** (Agricultural Collection Centre Management System)

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Executive Summary

The project delivered all 43 user stories of the Sprint 0 backlog as a live web system, deployed on 9 October 2026 and passing 72 automated tests. It replaces paper logbooks, WhatsApp ordering and per-centre spreadsheets with one platform for nine user roles, from farmer registration to buyer payment.

The team ran Scrum with one owner per epic and analysed its work against the eight performance domains and twelve principles of the PMBOK Guide, 7th edition [1]. Monitoring found that 25 of the 43 stories in the Sprint 0 proposal [2] needed rework before they met the Definition of Done, mostly for security and data-integrity defects; 10 stories had to be built new. Integration came late: 28 of the project's 36 commits landed in three days, 9 to 11 October 2026.

The main lessons are to baseline the backlog and Definition of Done before writing code, to integrate and deploy from the first sprint, and to keep every member's work visible in version control. Future work focuses on a mobile field app, scale and SMS integration, and a trained crop model once the system has collected real harvest data.

The live system can be reviewed as a web application and a REST API.

1. Introduction

The project replaced the paper, phone and WhatsApp operations of Saara Ketha Harvest Hub's collection centres with one web system that follows produce from farmer registration to buyer payment.

1.1 Background and business context

Sri Lanka loses an estimated 30 to 40% of its fruit and vegetables after harvest, about Rs. 20 billion a year, according to a National Audit Office figure reported to Parliament's Committee on Public Accounts [3]. Collection centres, such as the client's hubs [4], sit where much of that loss can be measured and reduced: they weigh, grade, store and resell produce from many smallholders.

Prices move every day. The Hector Kobbekaduwa Agrarian Research and Training Institute (HARTI) and the Central Bank of Sri Lanka (CBSL) publish daily wholesale prices for economic centres such as Dambulla and Peliyagoda [5], [6]. A centre that pays farmers and sells to buyers without current prices risks either underpaying farmers or selling at a loss.

1.2 The client

Saara Ketha Harvest Hub is a medium-scale agricultural collection and distribution enterprise. It links smallholder farmers with commercial buyers through collection hubs in several Sri Lankan districts [4]. Its centres run on paper logbooks, hand-written receipts, phone and WhatsApp ordering, and spreadsheets kept separately by each centre manager [4].

1.3 Problem statement

The client's manual processes cause six measurable business problems, identified in the Sprint 0 proposal [2] and confirmed in the group's architecture assessment [4].

Table 1: Business problems caused by the client's manual processes

Problem area	Current practice	Business impact
Farmer and crop records	Paper registers, re-entered at each visit	Duplicate or outdated farmer records; no verification before delivery
Weighing and grading	Hand-written weights and grades	Weight errors and grading disputes with farmers
Stock and expiry	No live view of batches or expiry dates	Spoilage and wastage noticed too late
Buyer orders	Phone and WhatsApp	Overselling and poor stock allocation
Farmer payments	Manual calculation from receipts	Slow, opaque payments that erode farmer trust
Management information	Separate spreadsheet per centre	No timely view of supply, demand, prices or farmer performance

1.4 Purpose and business idea

The purpose of the project was to design, build and hand over a secure, integrated web platform that digitises the full collection-centre process within the module's timeline. The business idea is a single system that connects every party in the chain and adds a multilingual crop advisor: farmers get crop suggestions for their district, and the centre gains early visibility of future supply [2].

The product vision from Sprint 0 framed the value: "an integrated digital platform that improves the management and traceability of agricultural produce" [2].

1.5 Objectives

The main objective was to develop a secure, integrated web-based Agricultural Collection Centre Management System within the project period that digitises farmer and crop management, collection and quality inspection, inventory and orders, financial processing, AI-based crop advisory and analytics [2]. Each sub-objective maps to one epic and was measured by accepted user stories against the Definition of Done.

Table 2: Objectives, measures of success and results

Objective	Measure of success	Result
Main: integrated system, live by the final evaluation	All four epics working end-to-end on a public deployment	Deployed on Vercel and Supabase on 9 October 2026
SO1: farmer, crop and multilingual AI advisory module (E1)	11 E1 stories accepted; advisor gives a suitability score with reasons in English, Sinhala or Tamil	11 of 11; score computed from district fit, crop rotation, soil pH and rainfall
SO2: digital collection, weighing and quality inspection (E2)	9 E2 stories accepted; net weight computed by the server; accepted plus rejected must equal net weight	9 of 9; rule enforced in the database and the form
SO3: inventory, buyer orders and distribution (E3)	14 E3 stories accepted; overselling and expired-stock allocation blocked	14 of 14; first-expiry-first-out reservation with a 70% reservation cap
SO4: payments, analytics and reporting (E4)	9 E4 stories accepted; payments from approved inspections and active prices; CSV and PDF export	9 of 9; three-month demand forecast and exports in place

1.6 Scope

The scope covered the four epics and 43 user stories of the Sprint 0 backlog [2], delivered as a responsive web application for nine roles: administrator, collection centre officer, farmer, quality inspector, inventory manager, buyer, finance officer, transport coordinator and manager. The team followed Scrum [7] and kept the backlog as the single source of scope.

Table 3: Project scope: what was in and out of scope

In scope	Out of scope (and why)
Farmer self-registration, verification, profiles, documents and crops	Native Android and iOS apps: the responsive web app covers phones for the prototype
Delivery appointments, weighing, grading, rejection reasons, evidence photos, QR batch codes and receipts	Live connection to digital weighing scales: weights are entered by the officer and validated
Inventory batches, warehouses, expiry tracking, orders, auctions and delivery scheduling	Online payment gateway: payments are recorded and receipted, not processed by card
Farmer and buyer payments, receipts, outstanding balances, management reports and exports	SMS alerts: notifications are in-app and, when configured, by email
Crop advisor bot, market price list for 55 Sri Lankan products, audit and complaint handling	A trained machine-learning model in production: no harvest history existed yet to train it

1.7 Solution overview

The solution uses three layers: a React single-page application [8], an Express REST API with role-based access control, and Supabase PostgreSQL with row-level security, authentication, storage and realtime [9]. Both the web application and the API are hosted on Vercel [10].

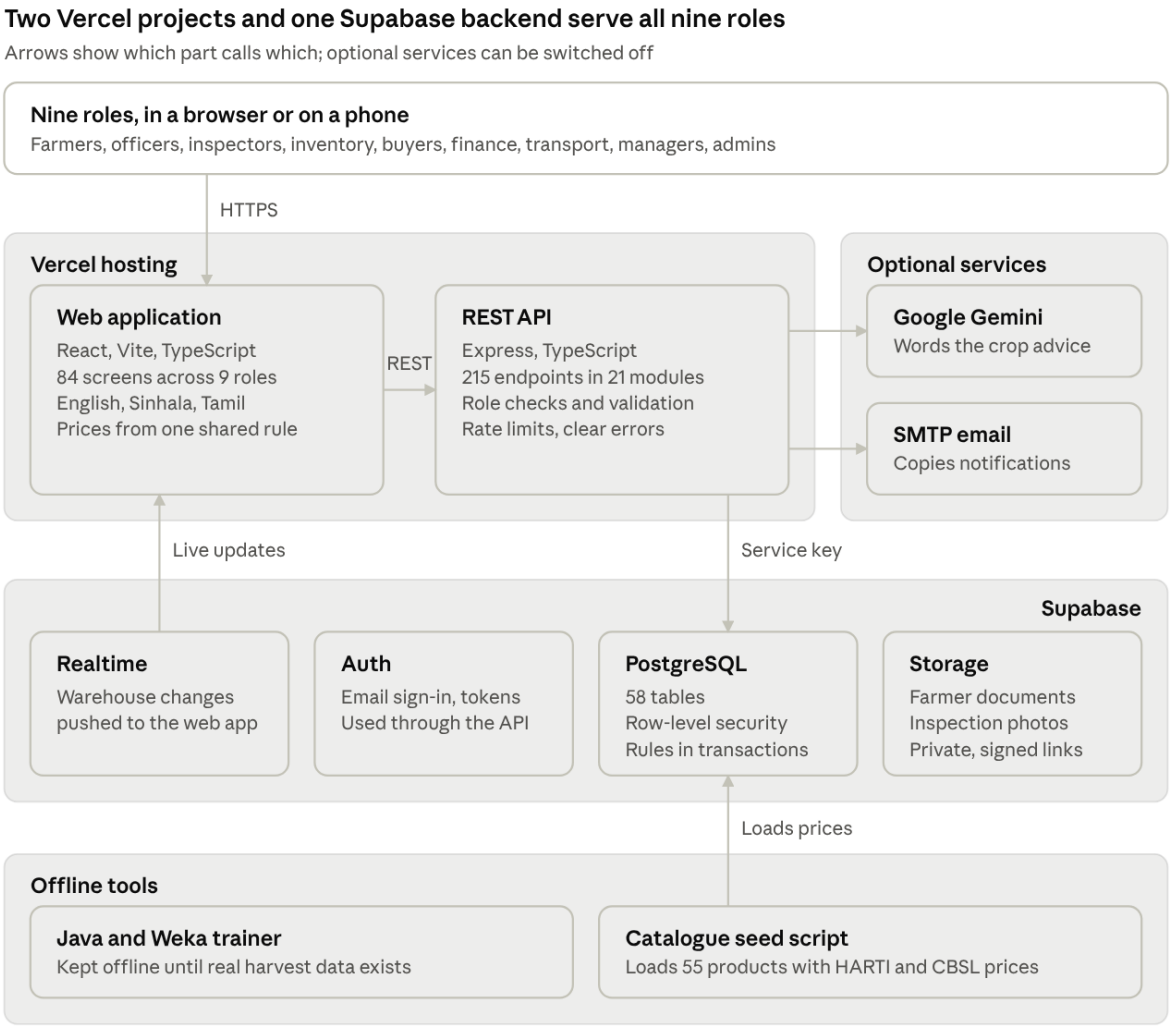


Figure 1: High-level system architecture of Saara Ketha Harvest Hub

Business rules run in the API and are enforced again in PostgreSQL. The web app talks to Supabase directly only for live warehouse updates, which row-level security still guards.

2. Project Execution, Monitoring and Control, and Closure

Execution ran in five phases, from a committed code baseline on 13 August 2026 to the showcase on 18 and 19 October 2026. Monitoring caught 25 stories that did not meet the Definition of Done and twelve defects before closure; all were corrected and the system was released.

2.1 Execution approach

The team used Scrum [7] with one owner per epic. The Scrum Master facilitated planning, reviews and the backlog, and each member owned one epic across its screens, API and database tables. Shared work (authentication, deployment, common components) was taken by whoever was free. Sprint 1 took the 21 highest-priority stories, at least half of each epic, as the proposal required [2].

Table 4: Execution phases, outputs and assessments

Phase	Dates	Goal	Main outputs	Assessment
Initiation and Sprint 0	to 29 Aug 2026	Agree problem, vision, epics, backlog and Definition of Done	Proposal: 4 epics, 43 stories, 195 story points, DoD, budget of LKR 95,500	Proposal presentation (10%) and report (15%)
Sprint 1: core functions	Sep 2026	Create, read, update and delete with validation for 21 stories	Registration, collection, inventory and payment basics	Sprint 1 evaluation (15%)
Sprint 2: integration	Sep to early Oct 2026	End-to-end workflows across the four epics	Inspection to batch to payment flow, order reservation, auctions	Sprint 2 evaluation (15%)
Stabilisation and release	9 to 11 Oct 2026	Every story meets the DoD; public deployment	28 commits; Vercel release; 10 new stories completed	Internal acceptance
Closure and showcase	18 to 19 Oct 2026	Hand over and present	55-product price list, A3 poster, 60-second video, this report	Final demo and viva (20%), report (25%)

2.2 Progress over time

The version-control record is the most objective progress evidence the project has. It shows a baseline committed before the proposal and a long quiet period, then a concentrated release push.

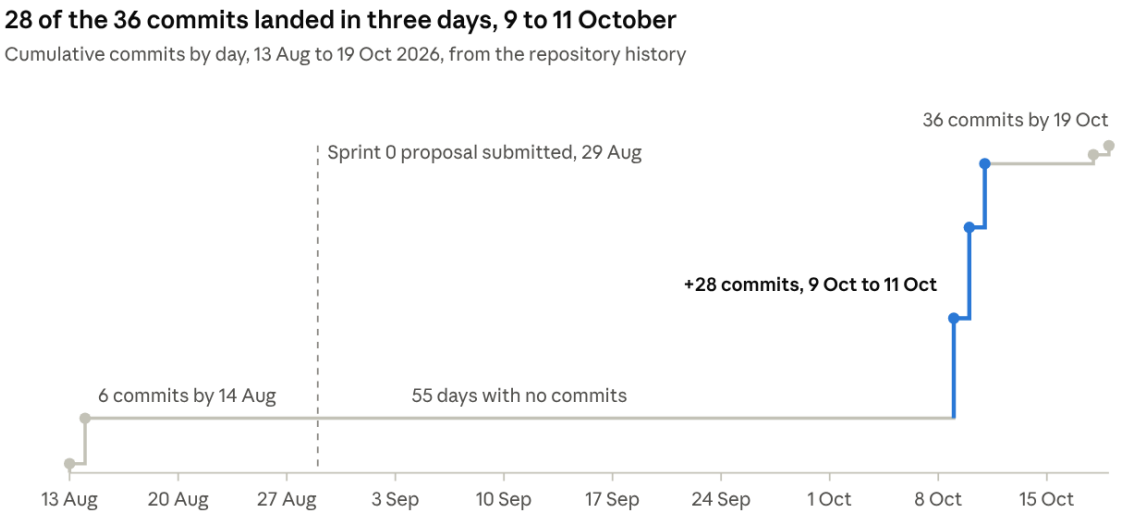


Figure 2: Cumulative commits by day, 13 August to 19 October 2026

Commits are an imperfect measure of effort, because all of them were pushed from one account. The shape is still clear: almost no work reached the shared repository between the proposal and the release week.

2.3 Monitoring and control activities

Seven control activities compared the work with the baseline and the Definition of Done. Each one produced evidence that is kept in the repository and summarised in the appendices.

Table 5: Monitoring and control activities and their evidence

Control activity	What was checked	When	Evidence
Story-coverage audit	Each of the 43 stories against the DoD; status marked Audited, Reworked or New	Before release	Appendix A
Automated tests	Auction rules, crop-advisor scoring, CSV export safety and error mapping, farmer validation	Every change	72 Jest tests in 4 suites; SQL tests for reservation, wastage and payments (Appendix D)
Type checking	Backend and frontend compile with no type errors	Every change	TypeScript compiler, no errors
Role-based API probes	Each role's endpoints called with a test session; test data removed and the removal verified	After each fix batch	Appendix D
Deployment checks	Build logs, login, CORS and environment variables on Vercel	Every deployment	Appendix E
Stakeholder review of the increment	Defects and requests raised by users testing the system became backlog items	9 to 11 Oct 2026	Change log (Section 2.4, Appendix B)
Data-source checks	Market prices compared with HARTI and CBSL bulletins	19 Oct 2026	Appendix G

2.4 Changes and change control

Twelve changes were accepted after the Sprint 0 baseline. Each became a backlog item and was accepted only if it served an existing epic objective; four requests that did not fit the time left were deferred to future work.

Table 6: Change requests accepted after the Sprint 0 baseline

ID	Change	Trigger	Impact	Decision
CR-01	Manager role with read-only reports and orders	E4-US6 to US9 name a Manager, but no such role existed	New role across database enum, routes, menus and translations	Accepted
CR-02	Crop advisor scores computed from four weighted factors, replacing fixed scores of 97, 92 and 87 by rank	Audit: scores were neither explainable nor testable	E1-US9 to US11 reworked; Java and Weka trainer kept offline	Accepted
CR-03	API and web app both hosted on Vercel instead of the planned Render or Railway	One platform, free tier, simpler configuration	Serverless entry point and CORS allow-list	Accepted
CR-04	Farmer documents, buyer feedback and email notifications	Named in epic descriptions but missing from the stories	Migration 027, three features	Accepted
CR-05	Complaint management for staff roles	Users found complaints could be raised but not resolved	New page and routes	Accepted

ID	Change	Trigger	Impact	Decision
CR-06	Near-expiry register with history	Request to keep a record of stock flagged near expiry	Migration 029	Accepted
CR-07	Product photos stored in the database; marketplace lists products with prices	User request	Migration 030 and image endpoint	Accepted
CR-08	Product-level prices editable by administrators and collection centre officers	User request	Price routes and validation	Accepted
CR-09	Driver NIC, phone and licence validation; warehouse pickup list; deliver action	Defects found in acceptance testing	E3-US8 reworked	Accepted
CR-10	Expected-harvest revisions tracked per crop	Farmers' expected harvests change during the season	Migration 031	Accepted
CR-11	Crop advisor filtered by district and redesigned	Usability feedback	E1-US10 reworked	Accepted
CR-12	Catalogue of 55 Sri Lankan products with grade A, B and C prices	Prices in the system were placeholders	Seed script and price list	Accepted

Deferred: online payment gateway, SMS alerts, native mobile apps and live scale integration (Section 6).

2.5 Issues and corrective actions

Twelve significant issues were found and closed. Six came from the code audit, three of them access-control and injection faults of the kinds listed in the OWASP Top 10 [11]; six appeared when the system was deployed and used.

Table 7: Significant issues and corrective actions

ID	Issue	Impact	Root cause	Corrective action
I-01	Anyone could self-register as an administrator	Critical security gap	Role taken from the request body	Self-registration limited to farmer and buyer; server rejects other roles
I-02	Farmer search filter accepted raw query text	Data exposure	User input placed directly in a query filter	Safe search and field whitelists
I-03	Farmers could change their own verification status	Bypass of E1-US4	Update accepted any field	Field whitelist on profile updates
I-04	Payments used invented grade-A inspections and fallback prices of Rs 200, 170 and 140	Wrong farmer payments	Placeholder code left in place	Payments only from approved inspections and the active price list
I-05	Inventory showed invented 50 kg batches	Stock overstated	Demo data in production code	Real batches only
I-06	Inspection photo upload called an API method that did not exist	E2-US6 broken	Contract drift between client and server	Method added and tested
I-07	Deployed web app called the local API address	Login failed in production	API address not set at build time	Build variable set in Vercel and redeployed
I-08	API rejected the hosted web app	Login failed in production	Environment values differed in case and domain	Values normalised; allow-list read from configuration
I-09	Vercel builds failed	No deployment	Lockfile generated on Windows; missing build tools	Install without the Windows lockfile; root build configuration
I-10	Days-to-expiry differed between local and hosted servers	Inconsistent stock risk	Hosted server runs in UTC	Dates computed in Sri Lanka time

ID	Issue	Impact	Root cause	Corrective action
I-11	Order subtotal ignored product, grade and centre	Buyers saw wrong prices	Price looked up by category only	One price function shared by web app and API
I-12	Users were logged out while navigating	Failed demonstrations	Login rate limiter counted the session check on every page load	Session checks excluded from the limiter

2.6 Risk management

The Sprint 0 register listed six risks [2]. Three of them materialised and were contained; three new risks emerged during execution and were added to the register.

Table 8: Risk register at closure

ID	Risk	Probability	Impact	Planned response	Outcome	Status
R1	Requirement changes	High	Medium	Regular requirement reviews	12 changes absorbed through the backlog	Closed
R2	Development delays	High	High	Sprint monitoring and task tracking	Materialised: 28 of 36 commits in 3 days	Closed; testing time was squeezed
R3	Data security	Medium	High	Authentication and role-based access	Materialised as I-01 to I-03; fixed before release	Closed
R4	AI and ML implementation	Medium	Medium	Phased implementation and testing	No training data; transparent weighted score used instead	Mitigated
R5	Database or service failure	Low	High	Backups and recovery	Managed database; migrations 001 to 031 can rebuild the schema	Open, monitored
R6	Integration issues	High	High	Continuous testing	Materialised: I-06 to I-11	Closed
R7	Configuration drift between local and hosted environments (new)	High	High	Not planned	Caused I-07, I-08 and I-10	Closed; settings documented
R8	Personal data leaking through exports and media (new)	Medium	High	Not planned	Readable data exports and showcase media kept out of version control	Closed
R9	Dependence on an external AI service (new)	Medium	Low	Not planned	Advisor works without it; the service only words the explanation	Open

2.7 Progress and status at closure

All 43 stories meet the Definition of Done at closure. The audit shows how much of that came from rework rather than first-time delivery.

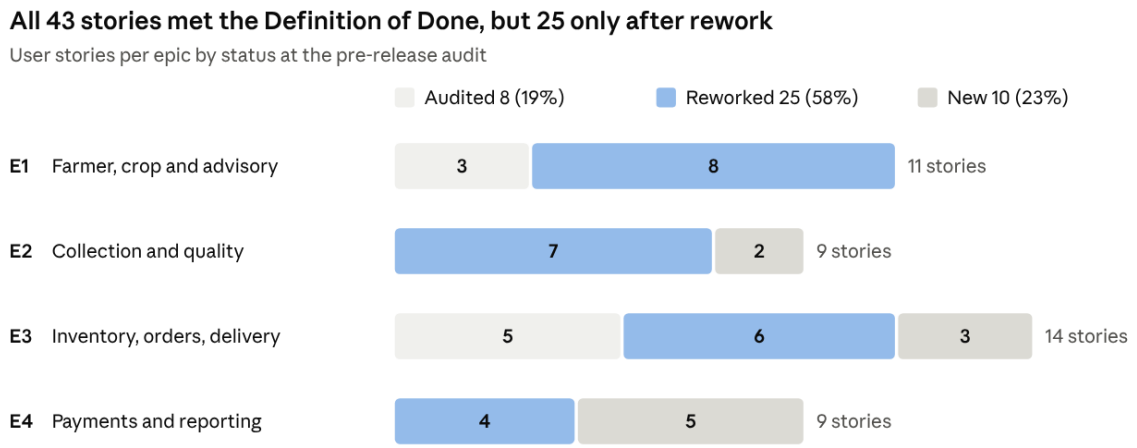


Figure 3: User stories per epic by audit status at closure

E1 had the highest share of rework, 8 of 11 stories, mostly in registration and the crop advisor. E4 needed the most new work: five of its nine stories, the receipts and management reports, did not exist before the audit.

2.8 Closure

Closure confirmed acceptance, released the system and handed over everything needed to run it. Two items are handed to the client to finish.

Table 9: Closure activities and status

Closure activity	Output	Status
Acceptance against the DoD	43 of 43 stories accepted (Appendix A)	Done
Release	Public web app and API on Vercel; database on Supabase	Done
Handover documentation	README, database migrations 001 to 031, setup scripts, per-epic study guides, price-list note	Done
Showcase	A3 poster and 60-second promotional video with narration and subtitles	Done
Lessons learned	Recorded in Section 5	Done
Open items for the client	Run the bidding and near-expiry setup scripts on the live database if not yet run; update the AI service model name; replace the published demo passwords before real use	Handed over
Final demonstration and viva	System demonstration and individual viva	Scheduled

3. Project Performance Domains Analysis

The project was strongest in Delivery and Stakeholders and weakest in Planning and Measurement. Each domain below is judged against the outcomes the PMBOK Guide sets for it [1], using evidence from the project.

Table 10: Summary assessment of the eight performance domains

Domain	Key evidence	Assessment	Main gap
Stakeholders	Every primary stakeholder became a system role; user feedback drove 7 of 12 changes	Strong	Feedback came late; no named Product Owner
Team	One owner per epic; Scrum Master role	Adequate	Nobody owned cross-epic integration; contributions not visible in version control
Development approach and life cycle	Hybrid: fixed assessment milestones, Scrum inside	Adequate	Hosted system not exercised until release week
Planning	Backlog of 43 estimated stories, DoD, budget, Sprint 1 plan	Weak	Inconsistent estimates; no dependency plan
Project work	Runbooks, setup scripts, study guides; zero-cost stack	Adequate	Manual database changes; defects reported without IDs
Delivery	43 of 43 stories accepted; live system	Strong	25 stories reworked; benefits not yet measured in use
Measurement	Story status, 72 tests, commit record	Weak	No velocity, burndown or estimate-versus-actual record
Uncertainty	Risk register; graceful fallbacks for AI and prices	Adequate	Register not updated during execution

3.1 Stakeholders

The project engaged its users directly through the product: each of the eight primary stakeholder groups in the Sprint 0 register [2] became a role with its own dashboard, plus a Manager role added later (CR-01). Farmers, the least technical group, were served in English, Sinhala and Tamil.

The weakness was timing. Most user feedback arrived after the system was deployed, so seven change requests (CR-05 to CR-11) landed in the last three days. The Sprint 0 role table also names no Product Owner, so the team set priorities itself instead of a client representative doing so.

Secondary stakeholders were reached only through data. Agricultural officers' knowledge entered through district crop tables, and market data through HARTI and CBSL bulletins [5], [6]; none were consulted directly, which limits confidence in the advisor's recommendations.

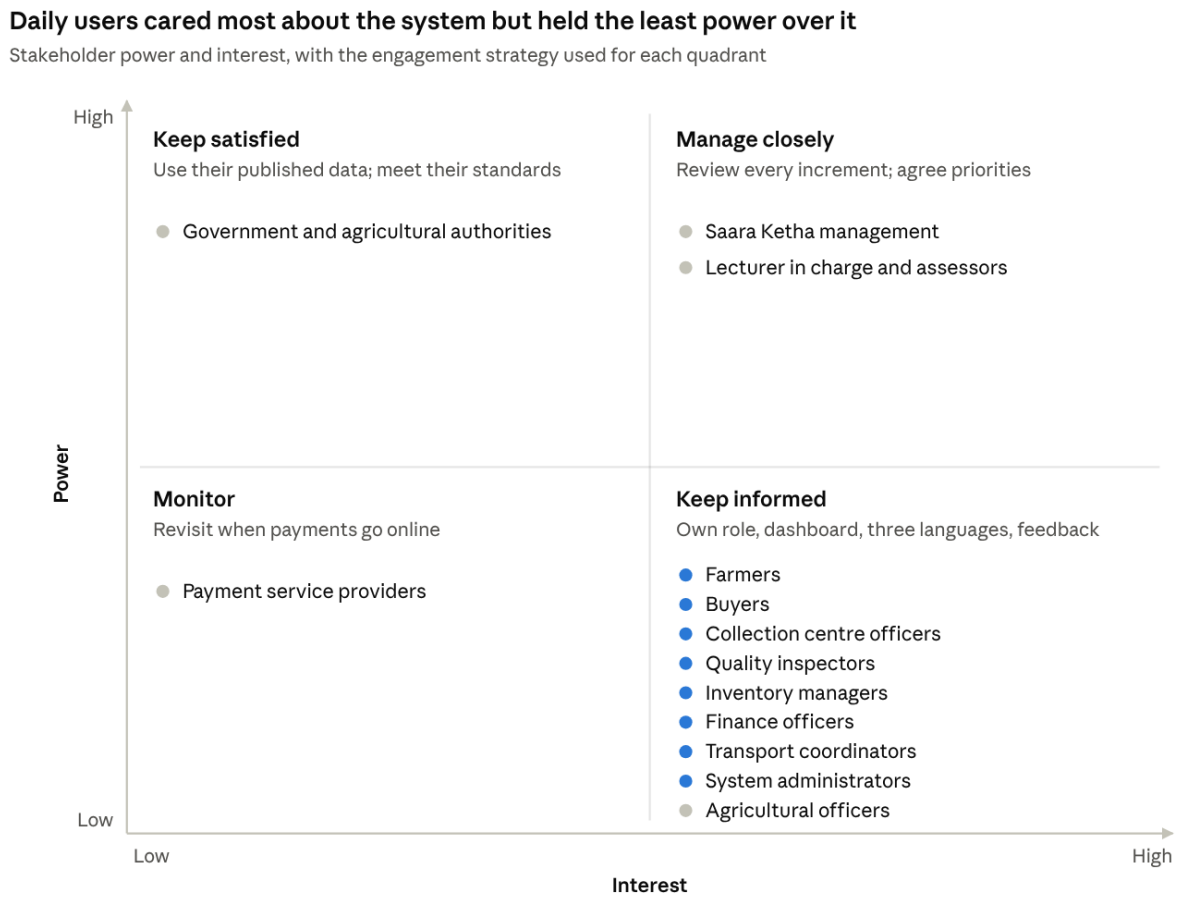


Figure 4: Stakeholder power and interest map with the engagement strategy for each quadrant

The eight groups that use the system every day (blue) sit in the low-power, high-interest quadrant. Moving their feedback earlier, through a Product Owner who speaks for them, is the main stakeholder improvement for the next project.

3.2 Team

Epic ownership gave every member a clear area of accountability, and the same allocation was reused in the group's architecture work [4], which kept a shared understanding across modules. The Scrum Master kept the backlog and the evaluation deadlines in view.

The gap was between the epics. The most valuable workflows cross epic boundaries: an inspection (E2) creates a batch (E3) that sets a farmer payment (E4). No one owned those joins, and the broken upload call (I-06) and the late integration push followed from that.

All 36 commits were made from a single account. Individual contributions are therefore not visible in version control, which weakens accountability, peer review and evidence for the individual viva.

3.3 Development approach and life cycle

The project used a hybrid approach, as recommended for most information-systems projects [12], [13]. The module fixed the milestones and deliverables in a predictive way, while the work inside each sprint was adaptive, with a re-prioritised backlog and working software reviewed at each evaluation [14].

Two life-cycle choices cost time. Each sprint was kept as a full copy of the code in its own folder, so fixes made in the main code had to be carried into three copies; tags or release branches would have avoided

this. A first hosting configuration was committed on 14 August, but the hosted system was not exercised again until the release week, so environment faults I-07 to I-10 surfaced then instead of in Sprint 1.

3.4 Planning

Sprint 0 produced a complete plan: 43 stories in the "As a, I want, so that" form [15], 195 story points with hour estimates and priorities, a Definition of Done, a Sprint 1 selection of 21 stories, hardware and software requirements and a budget [2].

The plan was internally inconsistent. Three stories carry different points in the backlog and the Sprint 1 plan (E1-US2, E3-US2 and E4-US3: 2 versus 3). The schedule appendix planned six sprints, but three were run plus a stabilisation phase. The budget appears as LKR 95,500 in the budget section and LKR 213,500 per year in the appendix without saying which is build cost and which is running cost.

The largest omission was dependency planning. Stories were estimated one by one, but the plan did not show that E2 inspections must exist before E3 batches and E4 payments can be tested. A dependency map with integration milestones would have pulled integration forward.

3.5 Project work

The team kept the work reproducible. The README gives a setup runbook, migrations 001 to 031 rebuild the database, setup scripts were handed to the database owner, and each epic has a study guide. The whole stack used open-source tools and free hosting tiers, so no procurement was needed; product photographs were sourced with licence credits recorded.

Two process gaps added effort. Database changes had to be run by hand in the Supabase editor, and at closure two scripts were still unconfirmed on the live database. Defects were reported as screenshots without IDs and were numbered only for this report; a tracked issue list from the first day would have given traceability.

3.6 Delivery

The project delivered its full scope: 43 of 43 stories meet the Definition of Done on a public deployment, and each proposal benefit now has a working capability behind it.

Table 11: Proposal benefits, delivered capabilities and measures after go-live

Proposal benefit [2]	Capability delivered	How to measure it after go-live
Improved data accuracy	Server-computed net weight; accepted plus rejected must equal net weight	Weight disputes per 100 collections
Faster farmer payments	Payments calculated from approved inspections and active prices, with approval and receipts	Days from collection to payout
Inventory visibility and less wastage	Live stock by crop and grade, expiry risk, write-off register	Kilograms written off per month
No overselling	First-expiry-first-out reservation with a 70% cap	Orders cancelled for lack of stock
Better traceability	QR batch codes linking farmer, inspection, batch and order	Share of batches traceable to a farmer
Better decisions	Management reports, forecasts, CSV and PDF exports	Report preparation time

Quality, however, was reached through rework rather than built in. 25 of the 43 stories (58%) failed the Definition of Done at the audit and had to be reworked. The benefits are also not yet measured, because the client has no baseline figures from the manual process.

3.7 Measurement

The project can show solid output measures: 43 accepted stories, 72 passing automated tests, 215 API endpoints, 84 screens, 58 database tables and a dated commit record. These describe what was built, not how efficiently or how predictably.

The repository holds no velocity, burndown or estimate-versus-actual record, so the 195-point estimate cannot be compared with the effort actually spent. Progress was judged by finished stories, which hides partial progress until late. Delivery measures such as deployment frequency and change failure rate [16] would have exposed the integration risk weeks earlier.

3.8 Uncertainty

The team turned two uncertainties into design decisions. With no harvest history to train a model, the advisor uses a transparent weighted score that farmers can follow; with prices changing daily, the price list keeps effective dates and a superseded history instead of overwriting values.

Other responses were reactive. The register in the repository is the Sprint 0 version, and the three new risks (R7 to R9) were recorded at closure, so their responses came after they had already caused issues. Reviewing the register at each sprint review would have flagged configuration drift before release week.

4. Application of the 12 Project Management Principles

Four of the twelve PMBOK principles [1] were applied well throughout the project. The other eight shaped real decisions too, but mostly in response to problems rather than by design, and that pattern explains most of the rework.

Table 12: Application of the twelve PMBOK principles

#	Principle	Strongest project example	Applied
1	Be a diligent, respectful and caring steward	Personal data kept out of version control; price estimates labelled honestly	Well
2	Create a collaborative team environment	One owner per epic; shared architecture work	Partly
3	Effectively engage with stakeholders	User feedback turned into seven changes	Partly
4	Focus on value	Sprint 1 covered at least half of every epic	Well
5	Recognise, evaluate and respond to system interactions	One price function shared by the web app and the API	Partly
6	Demonstrate leadership behaviours	Full audit of 43 stories before release	Partly
7	Tailor based on context	Hybrid approach and light, repository-based artefacts	Well
8	Build quality into processes and deliverables	Validation in form, API and database; 72 tests	Partly
9	Navigate complexity	Nine roles managed through layered, role-based design	Partly
10	Optimise risk responses	Fallbacks for the AI service and price history	Partly
11	Embrace adaptability and resiliency	Twelve changes absorbed; advisor redesigned	Well
12	Enable change to achieve the envisioned future state	Multilingual screens and receipts that build farmer trust	Partly

4.1 Stewardship

The team treated the client's data and money with care. Readable database exports and showcase media containing personal data were kept out of version control (R8), farmer identity documents sit in a private store behind signed links, and 14 of the 55 prices are openly marked as estimates rather than presented as official figures.

The stack cost nothing to license or host at prototype scale, well inside the LKR 95,500 budget [2]. One lapse remains: demo passwords are published in the project documents and must be replaced before real use, which is why it is listed as a closure item.

4.2 Collaborative team environment

Epic ownership gave each member a clear, respected area, and the team reused one shared model of the system across its project and architecture work [4]. This reduced arguments over who did what.

Collaboration stopped at the epic boundary. With every commit made from one account and no pull-request reviews, members did not see or check each other's code, and the joins between epics were nobody's job (Section 3.2).

4.3 Stakeholder engagement

The stakeholder register [2] was carried straight into the product: every primary group became a role with its own dashboard, and farmers can work in Sinhala or Tamil. Users testing the deployed system raised the defects behind CR-05 to CR-11, and each was acted on within days.

Engagement was reactive and late. Without a named Product Owner, priorities were set by the team, and users saw the integrated system only in the release week, when changes were most expensive.

4.4 Focus on value

Value drove scope decisions. Sprint 1 took the highest-priority stories covering at least half of each epic, so every stakeholder group got something usable early [2]. The payment gateway, SMS and native apps were deferred because a recorded payment with a receipt already gives the client most of the value.

The catalogue of 55 products with HARTI and CBSL prices (CR-12) turned the payment and ordering screens from a demonstration into something a centre could actually use. The missing piece is measured value: no baseline from the manual process exists yet (Section 3.6).

4.5 System interactions

Several fixes show systems thinking. Prices feed farmer payments (E4) and buyer orders (E3), so one shared price function now serves both the web app and the API (I-11). Inspection results and batch creation are written in a single database transaction, so a failure cannot leave half a record.

These interactions were mostly found through defects rather than mapped in advance. The broken upload call (I-06) and the time-zone mismatch (I-10) are both interaction faults that a simple context diagram, drawn at Sprint 0, would likely have exposed.

4.6 Leadership

The key leadership decision was to stop adding features and audit all 43 stories against the Definition of Done before release. That decision found the security faults I-01 to I-03 before any real user could exploit them.

Leadership was concentrated at the end and was mostly directive. Earlier, servant-leadership practices such as removing blockers at each sprint review would have spread the effort and reduced the release-week push.

4.7 Tailoring

The team tailored its method to an academic project with fixed evaluation dates: a hybrid life cycle, markdown artefacts kept beside the code, free hosting, and plain SQL scripts that the database owner could run without command-line tools [12], [13].

Tailoring was too aggressive in one direction. Dropping a burndown chart and an issue log saved little time but removed the early warnings the project most needed (Sections 3.7 and 3.5).

4.8 Quality

The delivered quality controls are strong. Business rules are enforced three times: in the form, in the API and in the database, as with accepted plus rejected quantity equalling the net weight. 72 automated tests

run in about one second, and missing records now return clear 404, 400 or 409 errors instead of server failures.

Quality was inspected in rather than built in. 25 of 43 stories (58%) failed the Definition of Done at the audit, which means the DoD was not applied when those stories were first called finished.

4.9 Complexity

The technical complexity of nine roles, 58 tables and 215 endpoints was contained by a layered design: 21 route modules, role checks on every route, and row-level security in the database [9]. Per-epic study guides made each part understandable on its own.

The social complexity was handled less well. The same team delivered this project alongside its Enterprise Architecture report, submitted on 14 October 2026 [4], and the clash of deadlines added to the October crunch.

4.10 Risk responses

Some responses were well judged. The crop advisor works even when the external AI service is unavailable (R9), prices are superseded rather than overwritten so an error can be traced, and stock reservations are capped at 70% to protect other buyers.

The planned responses to the two biggest risks were not scheduled. "Sprint monitoring" and "continuous testing" were written in the register [2] but had no owner or date, so delays (R2) and integration faults (R6) both materialised.

4.11 Adaptability and resiliency

The team adapted well. It absorbed twelve changes, replaced fixed advisor scores with an explainable model, and moved hosting to Vercel when that proved simpler. The system itself is resilient: it recovers cleanly from bad input, and the login limiter was tuned so normal navigation no longer logs users out (I-12).

The schedule was not resilient. With no buffer before the final evaluations, late changes cut into testing time.

4.12 Enabling change

The product is designed to change how the client works, not just to store data. Farmers register themselves, see their own receipts and payments in their own language, and can follow their payment status, which addresses the trust problem at the heart of the manual process.

The organisational side of that change was not planned. There is no training plan for centre staff, no plan to migrate the paper registers, and no parallel-run period; these are carried into future work (Section 6).

5. Lessons Learned

The project's most expensive lesson is that code written before the backlog and Definition of Done were agreed had to be rebuilt: 25 of 43 stories were reworked. The lessons below are drawn from the evidence in Sections 2 to 4 and each ends in an action the team will apply on its next project.

5.1 What worked

Table 13: Lessons learned: what worked

ID	Lesson	Evidence	Why it worked	Keep doing
W1	Audit every story against the DoD before release	I-01 to I-06 found and fixed before any real user saw them	A story-by-story checklist leaves nothing unchecked	Run the same audit at the end of every sprint, not only at release
W2	Enforce money and stock rules in the database, not only in the screens	Net-weight, 70% reservation and overpayment rules covered by SQL tests	No client, script or bug in the web app can bypass them	Make it the default for every rule that affects money or stock
W3	Prefer an explainable advisor when data is thin	Weighted score (district 40, rotation 30, soil pH 15, rainfall 15) shown with its breakdown	No harvest data existed to train a model; farmers can see why a crop is suggested	Choose explainable models until real data justifies a trained one
W4	Use official public data and label its source	41 of 55 prices from HARTI and CBSL bulletins, 14 marked as estimates	Credible to the client and to assessors; easy to update	Cite data sources inside the product
W5	Respond quickly to user feedback	Seven changes (CR-05 to CR-11) delivered in three days	Clear epic ownership and a modular codebase	Keep the speed, but collect the feedback in every sprint

5.2 What did not work

Table 14: Lessons learned: what did not work

ID	Lesson	Evidence	Cause	Impact	Action for the next project
L1	Building before the baseline creates rework	Code baseline committed 13 Aug; proposal and DoD submitted 29 Aug; 25 stories reworked	Development started before acceptance criteria and the DoD were agreed	58% of stories built twice; security faults shipped into early versions	Write no production code until the backlog, acceptance criteria and DoD are baselined; use throw-away spikes only
L2	Late integration and deployment concentrate risk	28 of 36 commits on 9 to 11 Oct; five deployment faults (I-07 to I-10, I-12) in release week	Hosting configured in August but not used again until release week; no owner for cross-epic flows	Testing time squeezed; failed logins during demonstrations	Deploy a thin end-to-end version in Sprint 1, deploy every sprint, rotate an integration owner
L3	Invisible individual work weakens the team	All 36 commits from one account; no pull-request reviews	Work combined and pushed from one account	No peer review; weak evidence of individual contribution	Every member commits from their own account through branches and reviewed pull requests
L4	Plan documents drifted apart	Three stories with two different estimates; six sprints planned, three run; two budget figures; sprint plan still shows Sprint 1 "In Progress" at closure	Artefacts written separately and never reconciled	Less trust in the plan; progress harder to measure	Keep one source of truth (a task board) and generate plan views from it; review it at each sprint

ID	Lesson	Evidence	Cause	Impact	Action for the next project
L5	Without progress data, slippage is invisible	No burndown, velocity or estimate-versus-actual record	Tailoring removed tracking to save time	Delays (R2) found only when deadlines arrived	Update a burndown and log actual hours per story each sprint
L6	Environment settings must be controlled like code	I-07, I-08 and I-10 all came from settings that differed between laptops and Vercel	Settings undocumented and set by hand	Production login failed twice	Commit a settings template, validate settings at start-up, test date logic in Sri Lanka time
L7	Manual database changes leave the live system uncertain	Setup scripts RUN_4 to RUN_9 run by hand; two unconfirmed at closure	No automated migration step	Live database may differ from the code	Apply migrations automatically from the pipeline on every deployment

5.3 Top three improvements

1. Baseline before building: agree acceptance criteria and the DoD in Sprint 0, and apply the DoD audit at every sprint review (L1, W1).
2. Integrate and deploy from Sprint 1, with one rotating integration owner and automated migrations (L2, L6, L7).
3. Make the work visible: individual commits with reviews, one task board, and a burndown updated each sprint (L3, L4, L5).

6. Future Work

The next release should close the gaps users meet at the collection counter: field use on phones, weighing, and messages for farmers without smartphones. Each item below comes from a scope exclusion, a lesson or a stakeholder need recorded in this report.

Table 15: Future work, priority and justification

Priority	Improvement	Limitation or need it addresses	Justification
Next release	Offline-capable mobile web app for officers and farmers	Rural connectivity is patchy; the current app needs a live connection	Reuses the React code, so it costs far less than separate Android and iOS apps
Next release	Digital scale and QR scanner integration	Weights are typed in by hand (Section 1.6); weight disputes were a core problem	Reading the scale directly removes transcription errors; QR batch codes already exist
Next release	SMS alerts in Sinhala and Tamil	Notifications are in-app and email only	Reaches farmers without smartphones or mobile data, the users most exposed to slow payments
Next release	Daily automatic import of HARTI and CBSL prices	Prices were loaded once, on 19 Oct 2026, but markets move daily [5], [6]	Keeps farmer payments and buyer prices fair without manual updates
Next release	Delivery pipeline with automated tests, migrations and preview deployments	Lessons L2, L6 and L7	Removes the configuration and database drift behind five release-week issues
Next release	Go-live change package: staff training, migration of paper registers, two-week parallel run	Principle 12 gap (Section 4.12)	Adoption fails if staff fall back to paper while records are incomplete
Next release	Benefit dashboard with baseline figures from the paper process	Benefits are not yet measured (Section 3.6)	Lets the client prove payment speed, wastage and dispute improvements
Later	Trained crop-recommendation model	The advisor uses weighted rules; the Java and Weka trainer is unused	Worth training once one or two seasons of harvest, inspection and price data exist, and only if it beats the rule-based score
Later	Payment disbursement through a bank or mobile-wallet provider	Payments are recorded and receipted but paid outside the system	Cuts payout time further; needs a provider agreement and security review
Later	Multi-centre load and failover testing	The client runs hubs in several districts [4]	Confirms the system holds up before every centre uses it at once

7. Conclusion

The project met its main objective: Saara Ketha Harvest Hub now has a live, integrated system covering all four epics and all 43 user stories, from farmer registration to buyer payment. Each sub-objective was met against its measure (Section 1.5), and the deployment, documentation and showcase were handed over at closure.

As project management, the result was sound but uneven. Stakeholder focus, value-driven scope and fast adaptation were real strengths, while planning consistency, measurement and early integration were weak, and 58% of stories had to be reworked. The PMBOK lens [1] shows a team that applied the principles mostly when problems forced it to.

The lessons are specific and actionable: baseline before building, integrate and deploy from the first sprint, and make every member's work and progress visible. With the future work in Section 6, the system is ready for a supervised pilot at one collection centre.

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Appendices

The appendices hold the evidence behind Sections 2 to 5. Each is referenced from the section it supports.

Table 16: Appendices and the report sections they support

Appendix	Content	Supports
A	Product backlog and story status at closure	Sections 2.3, 2.7, 3.6
B	Change and commit log	Sections 2.2, 2.4, 3.2, 5.2
C	Definition of Done verification	Sections 2.8, 4.8
D	Test and verification evidence	Sections 2.3, 4.8
E	Deployment and configuration evidence	Sections 2.3, 2.5, 3.5
F	Stakeholder register and engagement	Sections 3.1, 4.3
G	Showcase and data-source evidence	Sections 2.3, 2.8, 4.4

Appendix A – Product Backlog and Story Status at Closure

Story points are from the Sprint 0 backlog [2] and total 195. Status is the result of the story-coverage audit: Audited (met the DoD as built), Reworked (rebuilt to meet it) or New (built during stabilisation). All 43 stories met the DoD at closure.

Table 17: Product backlog and story status at closure

Story	Summary	Points	Audit status
E1-US1	Farmer creates own account	5	Reworked
E1-US2	Choose English, Sinhala or Tamil	2	Reworked
E1-US3	Officer registers and manages farmers	5	Reworked
E1-US4	Officer verifies farmer registrations	3	Reworked
E1-US5	Farmer views and updates profile	3	Reworked
E1-US6	Farmer registers crops	5	Audited
E1-US7	Farmer views crop history	3	Audited
E1-US8	Officer searches and filters farmers	3	Audited
E1-US9	Farmer uses the crop advisor bot	8	Reworked
E1-US10	Farmer gives location, previous crop, soil and climate	5	Reworked
E1-US11	Farmer sees recommendations with scores and reasons	8	Reworked
E2-US1	Farmer schedules a delivery	3	Reworked
E2-US2	Officer registers scheduled and walk-in deliveries	5	Reworked
E2-US3	Officer records gross and container weight	5	Reworked
E2-US4	Inspector inspects and grades produce	5	Reworked
E2-US5	Inspector records accepted and rejected quantities	5	Reworked
E2-US6	Inspector uploads evidence photos	3	Reworked
E2-US7	System generates batch numbers and QR codes	5	Reworked
E2-US8	Collection receipts and status	3	New
E2-US9	System validates weights, quantities and approval	5	New
E3-US1	Inventory batches	5	Reworked
E3-US2	Warehouses and storage locations	2	Audited

Story	Summary	Points	Audit status
E3-US3	Real-time stock levels	5	Reworked
E3-US4	Expiry and spoilage tracking	5	New
E3-US5	Buyer registers and browses products	3	Reworked
E3-US6	Buyer creates and tracks orders	5	Reworked
E3-US7	Order approval and stock reservation	5	Reworked
E3-US8	Delivery preparation and tracking	5	Reworked
E3-US9	Block overselling, expired stock and excess reservation	8	New
E3-US10	Buyer places bids	5	Audited
E3-US11	Buyer views own bids	2	Audited
E3-US12	Manager reviews bids	5	Audited
E3-US13	Manager accepts or rejects bids	5	Audited
E3-US14	Buyer is notified of bid result	3	New
E4-US1	Calculate farmer payment from quantity and grade	5	Reworked
E4-US2	Deductions and payment approval	5	Reworked
E4-US3	Farmer tracks payments	2	Reworked
E4-US4	Buyer payments and invoices	5	Reworked
E4-US5	Receipts and outstanding payments	3	New
E4-US6	Collection, inventory and quality reports	5	New
E4-US7	Farmer, buyer and waste analysis	5	New
E4-US8	Revenue, expenses, supply, demand and price trends	8	New
E4-US9	Forecasts and PDF or CSV export	5	New

Appendix B – Change and Commit Log

The repository holds 36 commits, all on one branch from one account, with no merge commits. The table lists them newest first with the change request (CR), issue (I) or risk (R) each one serves.

Table 18: Repository commits, newest first, with the linked change, issue or risk

Date	Commit	Change	Linked item
2026-10-19	630cb21	Sri Lankan catalogue of 55 products with market prices and photos	CR-12
2026-10-18	ffa5ab6	Stop rate-limiting the session check; keep showcase media out of version control	I-12, R8
2026-10-11	61e246b	Modal backdrop covers the whole window	UI defect
2026-10-11	07e9a09	Expected-harvest revisions in their own table	CR-10
2026-10-11	a1f4bbc	Crop advisor redesign	CR-11
2026-10-11	03a158f	Advisor offers previous crops that match the district	CR-11
2026-10-11	6a82a72	Register-crop link, centre contact validation, order price and subtotal	I-11
2026-10-11	cbaafe0	Edit product prices from the product cards	CR-08
2026-10-11	842ba6a	Product-level prices for administrators and centre officers, validated	CR-08
2026-10-10	544e52f	Licensed product photo seed	CR-07
2026-10-10	e2f9f3d	Product photos stored in the database and served by the API	CR-07
2026-10-10	5fe5ea3	Marketplace product view with prices	CR-07, CR-08
2026-10-10	e30e7e1	Near-expiry register; exported data kept out of version control	CR-06, R8

Date	Commit	Change	Linked item
2026-10-10	a0e63d8	Driver validation, warehouse pickup list, deliver action	CR-09
2026-10-10	f16ba49	Near-expiry days computed in Sri Lanka time	I-10
2026-10-10	51b070e	Menu labels fall back when a translation is missing	UI defect
2026-10-10	bb7b5ee	Farmer payments use the active price list	I-04
2026-10-10	c1e817d	Normalise environment name casing	I-08
2026-10-10	e1e76e7	Production defaults on Vercel; clear CORS rejection	I-08
2026-10-09	300ab40	Separate configuration errors from wrong passwords at login	I-07
2026-10-09	ec226f4	Clear 404 and 400 errors instead of server failures	DoD: error handling
2026-10-09	8702d89	Complaint handling for staff roles, notification links, CSV exports	CR-05
2026-10-09	2a547cb	Complaint management page	CR-05
2026-10-09	fe9d730	Complaints list, farmer auctions and profile access fixes	Defects
2026-10-09	a8ab15a	API and web app made safe for Vercel	I-09
2026-10-09	989e539	Install without the Windows lockfile	I-09
2026-10-09	c87220c	Root build configuration	I-09
2026-10-09	ecb97ec	Web app deployment configuration	I-09
2026-10-09	8ad2fc3	Configure Vercel deployments	CR-03
2026-10-09	aee507d	Audit rework synchronised: 206 files, 25,331 lines added, 3,319 removed	Appendix A, I-01 to I-06, CR-01, CR-02, CR-04
2026-08-14	d8052d1	Fix hosting configuration syntax	CR-03
2026-08-14	7957dc0	Hosting configuration	CR-03
2026-08-14	2057b9d	Hosting configuration	CR-03
2026-08-14	470fbf1	Root hosting configuration	CR-03
2026-08-14	17711ba	Floating crop-advisor button on the farmer dashboard	UI change
2026-08-13	e0120d4	Initial baseline of the full-stack platform	Baseline (L1)

Appendix C – Definition of Done Verification

Seven of the nine DoD criteria [2] were verified with evidence; two were only partly met and are carried into the lessons learned.

Table 19: Definition of Done criteria and how each was verified

DoD criterion	How it was verified	Result
Implemented to the story and its acceptance criteria	Story-by-story audit (Appendix A)	Met
Integrated without breaking other functions	End-to-end smoke scenarios (Appendix D)	Met
Validation, error handling, security and business rules in place	Server schemas, role checks on every route, database constraints and row-level security	Met
Data stored, updated and retrieved correctly	Role-based API probes with verified clean-up	Met
Interface complete, responsive and usable	Review at desktop and phone widths; promotional video recorded from the live interface	Met
Tested, with critical defects fixed	72 automated tests, SQL tests, issues I-01 to I-12 closed	Met
Code reviewed and follows agreed standards	Type checking passes; no peer review recorded in version control	Partly met (L3)

DoD criterion	How it was verified	Result
Documentation and story updates completed	README, story coverage, study guides, this report	Met
Approved by the Product Owner or a stakeholder	Accepted at module evaluations; no named Product Owner	Partly met (Section 3.1)

Appendix D – Test and Verification Evidence

All automated checks passed on 19 October 2026.

Table 20: Automated and manual verification results

Check	Scope	Result
Backend unit and API tests (Jest)	4 suites: auction rules, crop-advisor scoring, CSV export safety and error mapping, farmer registration validation	72 of 72 passed in about 1 second
Database tests (SQL)	Stock reservation and wastage; buyer payments and receipts	Passed on PostgreSQL 16
Type checking	Backend and frontend	No errors
Role-based API probes	Each role's endpoints called with a test session; test records deleted and the deletion verified	Passed after each fix batch

The five end-to-end smoke scenarios used for integration acceptance:

1. A farmer registers, an officer verifies them, the farmer books a delivery, the officer weighs it and an inspector grades it; an accepted plus rejected total that differs from the net weight is refused.
2. A buyer orders more than 70% of a crop's stock and is refused; an order within the limit is approved and reserved, and cancelling returns the stock.
3. Wastage is recorded on a batch and the expiry write-off runs; both appear in the wastage records and the manager report.
4. A partial buyer payment and then the balance produce two receipts, and the invoice becomes paid.
5. A manager logs in, management reports load and export to CSV, and no edit buttons appear.

Appendix E – Deployment and Configuration Evidence

The system runs as two Vercel projects over one Supabase database [9], [10]. Only setting names are listed; values are kept out of this report.

Table 21: Deployed components and required settings

Component	Location	Required settings
Web application	sara-ketha-harvest-hub-w8ld.vercel.app	VITE_API_BASE_URL (set at build time; its absence caused I-07)
REST API	sara-ketha-harvest-hub.vercel.app	SUPABASE_URL, SUPABASE_SERVICE_ROLE_KEY, SUPABASE_ANON_KEY, FRONTEND_URL, CRON_SECRET
Database	Supabase PostgreSQL	30 migrations, 001 to 031 (010 unused), then the setup scripts below

Table 22: Database setup scripts and their status at closure

Setup script	Purpose	Status at closure
RUN_4	Inspection evidence images	Run
RUN_5	Bidding status fix	To be confirmed on the live database
RUN_6	Readable views of the data	Optional; run when readable views are needed
RUN_7	Near-expiry register	To be confirmed on the live database
RUN_8	Product images	Run
RUN_9	Expected-harvest revisions	Run

System size at closure: 58 database tables, 215 API endpoints in 21 route modules, 84 screens and about 38,300 lines of TypeScript.

Appendix F – Stakeholder Register and Engagement

The register extends the Sprint 0 version [2] with how each group was engaged and served.

Table 23: Stakeholder register and engagement

Stakeholder	Type	Main interest	How the project engaged or served them
Farmers	Primary	Easy registration, fair grading, payment transparency, crop advice	Farmer role in three languages; receipts; crop advisor
Collection centre officers	Primary	Verification, deliveries, weighing	Officer role; centre phone and capacity validation added
Quality inspectors	Primary	Accurate grading and evidence	Inspector role with photos and rejection reasons
Inventory managers	Primary	Stock levels, expiry, allocation	Inventory role; near-expiry register (CR-06)
Buyers	Primary	Availability, orders, invoices, delivery	Marketplace with prices and photos (CR-07); auctions
Finance officers	Primary	Accurate payments and reports	Finance role; active price list in payments (I-04)
Transport coordinators	Primary	Scheduling and vehicles	Transport role; driver validation (CR-09)
System administrators	Primary	Users, configuration, security	Administrator role; complaint management (CR-05)
Collection centre management	Secondary	Performance, revenue, supply and demand	Manager role (CR-01) with reports and exports
Agricultural officers	Secondary	Accuracy of crop advice	Represented through district crop tables only
Payment service providers	Secondary	Reliable transactions	Deferred to future work
Government and agricultural authorities	Tertiary	Sector data and transparency	Their published price data used as the price source [5], [6]
Lecturer in charge and assessors	Academic	Module deliverables and standards	Sprint evaluations, showcase and this report

Appendix G – Showcase and Data-Source Evidence

The showcase files are kept outside version control because they show personal data from the system (R8); they are included in the submission package.

Table 24: Showcase deliverables and price data

Item	Description	File
A3 poster	System name, the four epics and the high-level architecture	docs/showcase/poster/poster-A3.pdf
Promotional video	60 seconds recorded from the live system, with narration and subtitles	docs/showcase/SaraKetha_Promo_voice_subtitles.mp4
Market price note	55 products (26 vegetables, 18 fruits, 11 rice) with purchase and selling prices per grade	docs/MARKET_PRICES.md

Price method: the purchase price is about 90% and the selling price about 115% of the median wholesale price; grade B is 85% and grade C 70% of grade A. Sources: HARTI for 35 products [5], CBSL for 6 [6], and 14 market estimates, each labelled in the system.